METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

DATASHEET E-203

E-203 · Final cooler (cooling water)

DOCUMENT NO.	:	MEOH-V1-ME-DS-005
DOCUMENT TITLE	:	Datasheet E-203
PROJECT	:	Methanol Converter Ver 1 · OTS
STATUS	:	Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

E-203 · Final cooler (cooling water)

Equipment list MEOH-V1-ME-EQL-001. Values from the simulator model at its design point unless marked assumed. P&ID: MEOH-PID-002.

General		
Service	Final cooler (cooling water)	MEOH-PID-002
Type	DEU / BEU: process (HP) in U-tubes, cooling water in the shell; horizontal	TEMA (assumed)
Shells	2 (in series)	

Performance (model, design point)		
Duty	132.9 MW	sensible 83.8 MW (model) + condensing 49.2 MW (Design tab, assumed latent heats)
Hot side in → out	155.9 → 40 °C	
Cold side in → out	32 → 42 °C	
LMTD / F	39.9 K / 0.966	2 shells in series, 2 tube passes each (one 1-2 shell would give F ≈ 0.78, below the usual 0.8 minimum)
Model UA	3451 kW/K	total duty; the model's sensible-only UA is 1980 kW/K (K.NTU203 × C_hot)
Service U (with fouling)	550 W/m²K	assumed, typical
Installed area	6910 m²	model UA + 10 % margin (assumed)

Design		
Tube side (process)	100 bar g / 225 °C	Zone B
Shell side (cooling water)	10 bar g / 225 °C	assumed; design T = process side (CW can be blocked in)
CW flow	11433 t/h (32 → 42 °C)	total duty, assumed return 42 °C
Code	ASME VIII Div. 1 (assumed); TEMA R for exchangers	
Corrosion allowance	3 mm (CS / Cr-Mo parts), 0 (tubes)	assumed

Construction		
Tubes	9464 × OD 19.05 × 2.11 mm × 12.2 m (4732 per shell)	triangular pitch 23.8 mm
Shell inside diameter	1850 mm	tube count by TEMA bundle factors
Materials	1¼Cr-½Mo SA-387 Gr 11 Cl 2 channel; tubes duplex 2205; shell (CW) SA-516 Gr 70	EQL §3
Thickness (pressure side)	channel 65.8 + 3 → 70 mm (ID 1850)	UG-27(c)(1)

Notes		
1	Tube rupture: HP gas into the 10 bar g CW side — relief required on the CW side (RLB, EQ-er).	
2	The simulator cooler counts the sensible heat only (EQ-em-02).	
3	TI-254 on the CW return; TSV-203 thermal relief on the CW side (Safety layers tab).	